

INF222 - Practical Work Submission

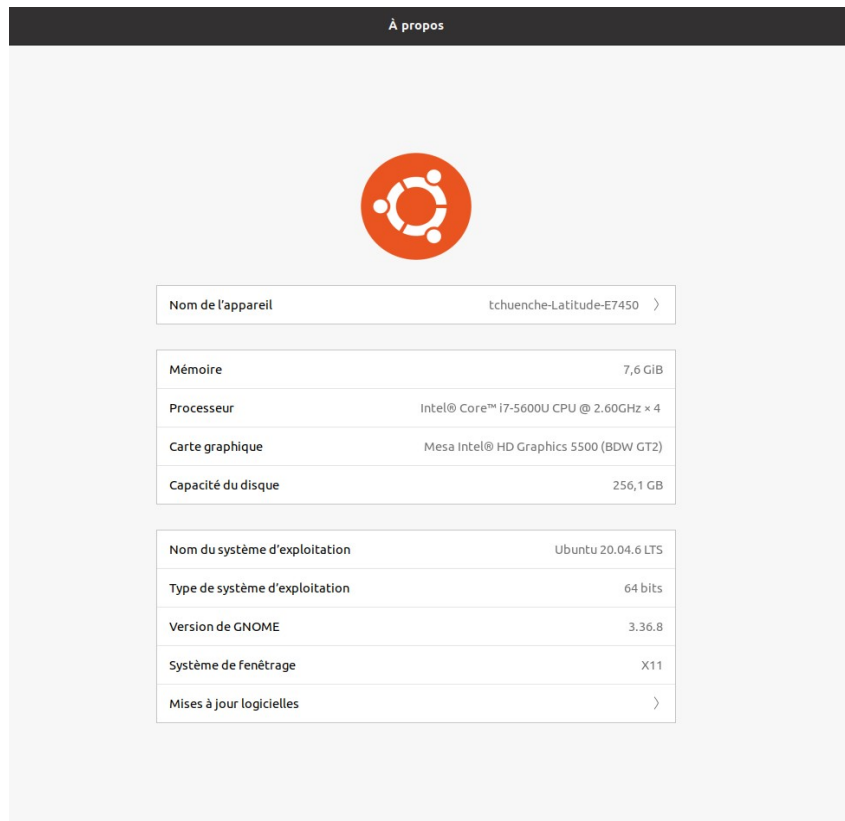
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INF2222 : Backend Development

Exercice_0 : Setting up development enviroment :

I am using Ubuntu 20.04.6 LTS for this project, leveraging its robust environment for backend development



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Technology/Tool	Programming Language	Purpose	Specific Version/Proposed Tool
Backend Language	Python	As the main programming language	python:3.8-slim
Web Framework	Flask	For the backend development of the application	flask==3.0.3

Database System	PostgreSQL	To manage the database of the app	postgres:13-alpine
Code Editor/IDE	VS code	My development enviroment where the app is developed	Vs code 1.101
Docker	Docker io	The Docker where the app is Dockerise for deployment	Docker version 24.0.5
Version Control	Git	To manege my versions	git 2.49.0

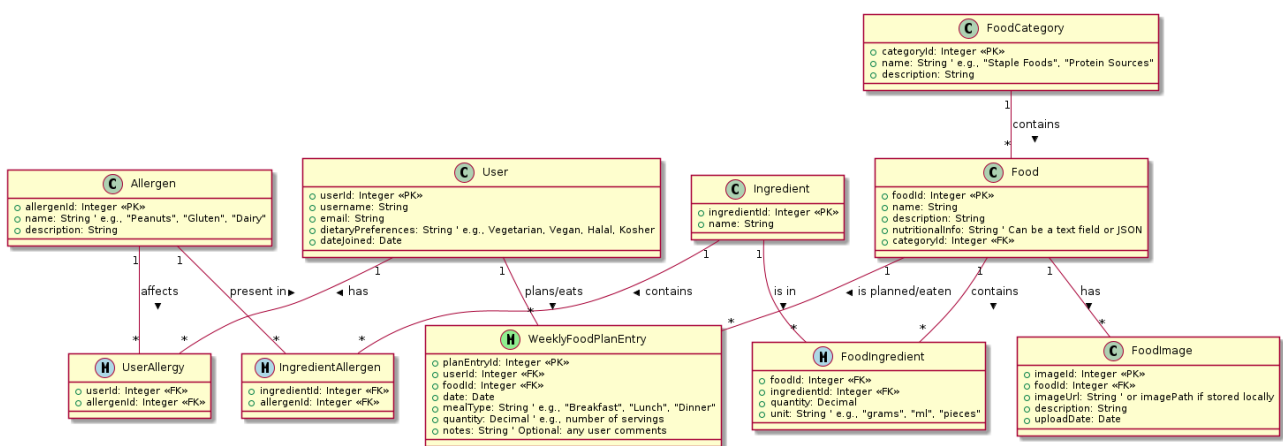
3- Install the development environment

- python3 : `sudo apt install python3 python3-pip`
- Flask : `pip install flask flask-sqlalchemy pycpg2-binary`
- PostgreSQL : `sudo apt install postgresql postgresql-contrib`
- Docker : `sudo apt install docker.io docker-compose`

Exercise_1 : Data collection.

1- I downloaded differnt data images and store it in my FOOD_APP/static which i can acces easily

2- I proposed a complete and robust UML diagram for my food app.



Exercise_2 : In this exercise, i implemented the DAO modules using **psycpg2** as my connectivity API and later on i used **sqlalchemy** for the ORM approach. Where i build and populated my databases.

Exercise_3 : Business logic implementation

The business module encompasses the core functionalities and entities that define the application's domain.

a) Module for Weekly Food Planning

This module is designed to empower users to organize and schedule their food consumption on a weekly basis, linking specific food items to particular dates

and meal types. Definition of a WeeklyFoodPlanEntrySchema using ma.SQLAlchemyAutoSchema for data serialization/deserialization.

b) Module for Managing a Buffet in the Case of a Ceremony

This module is conceptually designed to assist in the management of food items for events or buffets, facilitating tracking of food prepared and consumed.

Exercice_5 : Deployment, Design and Implementation of the API Endpoint

Here, i used the Docker technologies to Dockerise my backend application. Secondly i did a file that contains all my API endpoints for all the services my app provide. (the file is in the same folder as this one named : « API_ENDPOINT.pdf»)